

Gaze Sampling of Motion Relates Differently to Authenticity Judgments for Real and AI-Generated Physics Videos

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Authenticity judgments about AI-generated video may depend on how viewers sample motion over time, but broad gaze summaries collapse across locations and moments. They therefore cannot reveal whether viewers look toward motion-rich regions when those motions occur. We examined whether video-level temporal organization and moment-by-moment gaze–motion alignment were associated with judgments of real versus AI-generated physics clips. We conducted a secondary analysis of the physics-video subset of a previously published eye-tracking dataset, comprising 20 matched real–AI scene pairs and 385 eligible judgment trials from 40 participants. For each clip, we quantified temporal irregularity using connected-component persistence (video H_0 total persistence) in a per-clip normalized temporal-irregularity field. We also calculated an eccentricity-controlled gaze–motion coupling score by matching gaze and motion within the same video and time point. AI clips had lower video H_0 total persistence than matched real clips. Gaze–motion coupling was associated with judgments in opposite directions across clip classes: greater coupling corresponded to lower odds of an AI judgment for real clips and higher odds for AI-generated clips. This class-dependent pattern suggests that video temporal structure and moment-specific gaze sampling provide complementary information about authenticity judgments; gaze–motion alignment alone is not a uniform cue across real and AI-generated clips. Because this is a secondary analysis of a physics-video subset and T2 judgments, the results should be interpreted within that stimulus and task domain. These associations are descriptive and do not establish that coupling caused judgments.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Computer vision*.

Additional Key Words and Phrases: eye tracking, dynamic scene viewing, generative video, visual attention, gaze–motion coupling, persistent homology

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1 Introduction

Generated videos of physical events raise distinct questions about visual appearance, temporal structure, and physical consistency. Benchmark work on generative video has treated physical principles as an evaluation target separate from visual realism, motivating measurements that describe temporal image structure without treating those measurements as direct ratings of plausibility [6]. Viewers, meanwhile, make authenticity judgments while sampling only part of a changing scene. Eye tracking can characterize where and when that sampling occurs, but a final real-or-AI response does not by itself identify the visual information encountered during viewing.

Across 23 comparative studies, a recent review found no modality-independent gaze signature of AI-generated content and called for clearer links between gaze measures and functional outcomes [8]. In an image-based authenticity

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task, participants sometimes judged photographs of real faces as AI-generated, illustrating why judged and actual authenticity should be kept distinct [4]. In a separate text-to-video study without eye tracking, longer viewing improved detection of generated clips while accuracy for authentic clips remained approximately constant; participants reported using motion among other judgment cues [2]. Shi et al. recorded eye movements from 40 participants viewing real and AI-generated videos in a video-understanding task (T1) and an AI-detection task (T2) [9]. Their analysis reported that the broad gaze measures considered did not differ reliably by a clip’s actual authenticity, while task framing and participants’ perceived authenticity were related to gaze behavior. The present article reanalyzes the same participant data and source experiment, selecting only the physics category (Set1/S1). It therefore complements the published study. We ask whether measures derived from processed video and gaze fields reveal conditional relationships that were not addressed by the earlier broad gaze summaries.

Motion is one feature that can organize gaze during dynamic-scene viewing, although its influence depends on the scene and task [5]. During viewing of unedited real-world videos, task instructions shifted gaze away from high-flicker regions and people after initially center- and motion-biased saccades [10]. In a dynamic body-expression task, gaze allocation varied with the informativeness of facial and movement cues [7]. Gaze distributions in natural dynamic scenes can also reflect center-related and composition-related biases [12]. A raw average of motion at fixation locations may therefore conflate motion sampling with where viewers tend to look. We use a within-video control that matches gaze locations by radial eccentricity and samples them at the same moments as observed fixations. The resulting score describes the relative motion magnitude at sampled gaze positions; it does not measure eye velocity aligned with optical flow or identify attention to a verified anomaly.

We also summarize temporal structure with zero-dimensional persistent homology. Across a sequence of thresholds, connected regions appear and merge; their birth and death values provide a multiscale description of connectivity [1]. Here the calculation is applied to a three-dimensional, per-clip normalized field of forward–backward flow and photometric inconsistency, and H_0 total persistence sums finite connected-component lifetimes. It is an algorithmic descriptor of the processed field, not a semantic map of artifacts or violations of physical laws. Prior topological work on gaze has addressed raw-trajectory segmentation [3]; our gaze measure instead applies cubical persistence to a smoothed, duration-weighted gaze-density field [11].

The analyses address three questions. First, do the 20 matched physics-scene pairs differ in video H_0 total persistence? Second, within eligible T2 trials, is controlled gaze–motion coupling associated with an AI judgment differently for actual-real and actual-AI clips? Third, does video H_0 covary with gaze H_0 across the four actual-authenticity-by-judgment cells, and is that result stable to zero-safe outcome handling? The paired video comparison uses scene pairs as its independent units; the judgment and gaze analyses use repeated trials with participant, scene, and video structure represented in the models.

2 Related Work

3 Methodology

3.1 Overview and interpretation of the measures

This secondary analysis asks how the visual structure of short physics videos relates to the way people sample them with their eyes, and whether that sampling is associated with recognizing AI-generated clips. The analysis has four connected parts. First, it compares the visual structure of 20 AI videos with that of 20 real videos showing matched physical scenarios. Second, it asks whether a participant looked at stronger-motion locations than would be expected for

gaze points at the same distance from screen center and at the same moments. Third, it tests whether that gaze–motion measure is associated with an AI judgment, especially a correct AI judgment on an AI clip, after accounting for simpler gaze summaries. Finally, another analysis asks whether the video’s topological structure and the structure of the participant’s gaze field covary differently across the four combinations of actual and judged authenticity. The stimulus comparison uses matched video pairs as its units; the behavioral models use eligible viewing trials and account for repeated participants and stimuli.

Supplementary Figure S1 illustrates the actual cubical filtration, persistence diagrams, and interval barcodes used to make the topology measures concrete.

How to read the video-topology measure. We describe each five-second video as a three-dimensional grid: horizontal position, vertical position, and time. At every grid location we estimate temporal irregularity from disagreement between forward and backward motion estimates and from changes in image intensity that remain after motion-based warping. These values are measurements of temporal structure, not labels for AI artifacts or violations of physics. We gradually change a threshold on the grid and record when connected regions appear and merge. Zero-dimensional persistent homology, denoted H_0 , summarizes these connected regions; total H_0 persistence adds their lifetimes across threshold values. This is analogous to examining a landscape at many water levels and asking how long distinct islands remain separate. Here the landscape includes time as well as space. A smaller total does not mean a clip contains fewer visible objects, that any particular event lasted less time, or that observers noticed a specific flaw.

How to read direct gaze–motion coupling. For every fixation, we measure the motion strength at the point and time viewed. Looking toward the center of a video can already produce a high score because center-screen locations often carry motion and the preceding fixation display encourages central gaze. We therefore compare each observed gaze position with 30 control positions that have the same radial distance from the center, lie on valid video support, and are sampled at the same video times. We calculate a trial-level score in units of the control-score standard deviation. A positive score means the observer sampled more motion than expected under this spatial and temporal control. It measures *where and when* gaze fell relative to motion strength; it does not measure whether the eyes moved in the direction of optical flow, whether gaze tracked one physical object continuously, or whether the motion was anomalous.

How to read gaze topology and topology alignment. Fixations were also converted into a smooth, duration-weighted field on the video’s space–time grid. Its H_0 summary describes the multiscale organization of regions receiving gaze mass. We kept this gaze-field representation separate from conventional measures such as scanpath length and dispersion. One comparison score asks whether a trial’s gaze persistence landscape is closer to the viewed video’s landscape than to those of 10 other physics videos with the same actual authenticity. This comparison concerns topological *shape*; it does not preserve absolute image location and cannot show that eyes landed on a specific irregularity. The video–gaze model asks a different question: whether clips with larger video H_0 values tend to elicit larger gaze H_0 values, conditional on actual authenticity and the participant’s judgment. These two topology analyses should not be conflated.

Interpretation of the statistical models. An AI judgment is a participant’s response, whereas correct AI detection refers only to actual-AI trials judged AI. The main all-T2 judgment model estimates whether the coupling–judgment slope differs between actual-AI and real clips and gives the slope within each actual class. A supplementary AI-only fit and nested model comparison check the narrower detection question. None of these observational associations identifies a causal effect of gaze behavior, and a fitted-likelihood comparison is not an estimate of out-of-sample predictive gain.

3.2 Participants, stimulus subset, and procedure

We reanalyzed the original eye-tracking study of Shi and colleagues [9], restricting its stimulus library to the physics set (S1). The original sample comprised 40 adults with normal or corrected-to-normal vision (26 women; age range 18–48 years). The source study was approved by the Department of Engineering Ethics Committee at the University of Cambridge, and all participants gave informed consent [9]. The present work used de-identified fixation and response records; it did not recruit new participants.

S1 comprised 20 real videos drawn from Physics-IQ [6] and 20 AI-generated videos depicting the corresponding physical scenarios. The scenes covered solid mechanics, fluid dynamics, optics, thermodynamics, and magnetism. The AI version of each scene was generated from a description of the physical event and a selected keyframe of its real counterpart. Real and AI clips were cropped to 960 by 720 pixels, trimmed to five seconds, and presented without generation watermarks. Pairing therefore controlled scenario identity but did not make corresponding frames pixel registered. The present stimulus analysis used all 40 clips and defined 20 AI–real scene pairs.

The source experiment had two consecutive phases. In T1, participants watched a clip and described the event in one sentence. After a break, in T2, they watched a different set of clips and judged each as real or AI-generated, with spoken responses transcribed by the experimenter. Each participant viewed 20 different clips in each phase, half real and half AI across the full stimulus library. A central fixation display preceded the next video. Counterbalancing prevented any participant from seeing both versions of one scene or seeing the same scene in both phases. T1 always came before T2, so phase cannot be separated from order, practice, or fatigue. T1 records contributed to eligibility accounting and full-table covariate standardization; all judgment and the perception-aware models reported here used T2 only.

Gaze was recorded at 60 Hz with a Gazepoint GP3. The screen resolution was 1680 by 1050 pixels. The working presentation rectangle for this analysis placed the 4:3 video in the aspect-fit area from (140, 0) to (1540, 1050) display pixels. Its displayed width was 1400 pixels and its height was 1050 pixels. These geometry values and the original-study interpretation of the timing field were fixed before gaze-to-video registration in the frozen run.

3.3 Frozen inputs, response reconciliation, and analytic units

The frozen S1 run drew fixation, judgment, and presentation-list tables from separately retained input files and used the 40 local S1 source MP4s. It matched each presentation slot to an explicit video identifier. Actual authenticity was determined from the final `-a` or `-r` video-ID suffix; scene identity was the same identifier without that suffix. We did not use an authenticity field embedded in an earlier aggregate fixation file, because that aggregate had a positional-label problem. The reconciliation script required a unique mapping from study, counterbalancing group, task, and presentation slot to video; a unique response on the full trial key; and agreement between the response and video tables on the assigned video. Fixation and response records were joined on Project, UserID, Task, MEDIA_NAME, and VideoID within the original study. The script checked that correctness equaled agreement between actual and judged authenticity. T1 carried no binary judgment outcome.

A *trial* was one participant’s viewing of one clip in one phase. A *scene pair* comprised two different clips, one real and one AI, portraying the same scenario. In the paired stimulus test, the independent observations were the 20 scene-level AI-minus-real differences. In the participant models, each eligible trial contributed one gaze score and, for T2, one judgment. Participant, scene, and video identifiers were retained separately because viewers contributed multiple trials and each stimulus was viewed by multiple people. The run preserved source tables, reconciled trial keys, video manifests, feature arrays, model tables, and stage-specific provenance files.

3.4 Registering fixations to the five-second video

The original-study Time value was treated, as author-verified, as the fixation's end time relative to video onset. If e_k is Time and d_k is the recorded duration FPOGD, the raw interval of fixation k was $[e_k - d_k, e_k]$. We intersected it with the displayed video window $[0, 5]$ seconds. Thus a fixation beginning at or after 5 s had zero contribution, and one crossing 0 or 5 s contributed only its within-window overlap. Fixation timing was not inferred from its order or from the Gazepoint system clock. The script required consecutive fixation indices within a trial and quarantined a trial when finite fixation endpoints were nonmonotone or adjacent recorded intervals overlapped beyond a 0.0001-s tolerance. A recorded fixation longer than 2 s was flagged for quality review; this flag alone was not an exclusion rule.

Gazepoint FPOGX and FPOGY expressed position as fractions of the entire screen, with the origin in the upper left. We first converted to display pixels, then to video-normalized coordinates. The video and gaze coordinate systems both increased downward in y , so no vertical flip was applied. We required $0 \leq x_k, y_k < 1$; gaze outside the displayed video was excluded, never snapped to the nearest image edge. For distance-based summaries, video-normalized coordinates were converted to source-video pixels by multiplying x by 960 and y by 720.

An eligible fixation had finite position, end time, and duration; positive recorded duration; tracker validity FPOGV=1; positive overlap with the five-second video; an in-video coordinate; and a trial that passed the interval-order check. A trial entered the analytic model table only if at least five such fixations provided at least 2.5 s of cumulative in-window duration. These criteria were applied after timing, validity, and viewport exclusions. Of 787 candidate physics-video gaze trials, 769 passed: 384 T1 and 385 T2. The retained T2 set included 193 actual-AI trials and 192 real trials from all 40 participants. Fixations failing the rules, along with their reasons, remained in quality-control outputs.

3.5 Video standardization and common feature grid

Source clips differed in encoded frame timing, so feature extraction used presentation time rather than source frame index. The manifest script checked that filenames matched the declared 40-video library, that every scene had one AI and one real member, and that decoded timestamps increased monotonically. It recorded codec metadata, duration, first and last presentation timestamps, and SHA-256 hashes. The standardization script rejected changes to source hashes or unsupported differences from the declared five-second window. It used FFmpeg to clone the last frame for at most 0.06 s if an otherwise acceptable encode ended slightly early, trim to five seconds, resample to 24 frames per second, resize with Lanczos interpolation to 320 by 240 pixels, convert to grayscale, reset presentation timestamps, and write a lossless FFV1 derivative. It checked that each derivative had exactly 120 frames at the target size and rate.

All downstream video and gaze fields shared a grid with 60 temporal bins and 60×80 spatial bins. A temporal bin spanned $5/60 = 0.0833$ s; a spatial bin represented a 4 by 4-pixel block of the 320 by 240 derivative. The 119 interframe maps of each clip were assigned by their midpoint timestamps to the 60 temporal bins. Values in each space-time cell were pooled across its contributing frame pairs and pixels using the mean of finite measurements. This shared grid made trial gaze fields and video fields comparable in scale and prevented the video's original encoding rate from determining the number of measurements in a topological summary.

3.6 Motion and temporal-irregularity fields

Two fields were derived from each video. The motion field M measured the magnitude of image displacement after an estimate of global camera movement was removed. The temporal-irregularity field A combined failures of forward-backward flow consistency and of photometric consistency after warping. Motion and temporal irregularity can both arise in ordinary real footage; neither was validated as a local AI-artifact map.

For each adjacent frame pair, dense forward and backward optical flow were estimated with the Farneback algorithm using a pyramid scale of 0.5, three levels, a 15-pixel window, three iterations, polynomial neighborhood size five, and polynomial smoothing 1.2. The camera-motion component was estimated from up to 1000 ORB features: binary descriptors were matched with cross-checking, the 300 best matches were retained, and a homography was fitted with RANSAC at a three-pixel reprojection threshold. The routine required at least eight matches and six inliers. If estimation failed for a frame pair, the configured policy used uncompensated flow for that pair and logged the failure. The local-motion magnitude was the Euclidean norm of forward flow minus the homography-derived displacement, multiplied by 24 to express motion in derivative pixels per second.

The consistency calculations used the *total* forward-flow coordinate map, even when the motion magnitude used camera compensation. At image location $\mathbf{x}$, let $\mathbf{f}(\mathbf{x})$ be forward flow and $\mathbf{b}(\mathbf{x} + \mathbf{f}(\mathbf{x}))$ be interpolated backward flow. Forward-backward disagreement was $\|\mathbf{f}(\mathbf{x}) + \mathbf{b}(\mathbf{x} + \mathbf{f}(\mathbf{x}))\|$. Photometric disagreement was the absolute gray-level difference between the earlier frame at $\mathbf{x}$ and the later frame sampled at $\mathbf{x} + \mathbf{f}(\mathbf{x})$. Warps leaving the later frame were marked invalid rather than treated as high-error image edges. Pooled irregularity voxels required at least 50% finite transport support.

Within each video, the pooled motion volume was clipped and scaled to $[0, 1]$ using its first and 99th percentiles. The two irregularity components were individually scaled in the same way over supported voxels; their equally weighted mean was scaled again with the first and 99th percentiles to form A . Unsupported A voxels were filled with zero for storage, and a separate validity mask was saved. We also retained the mean *unscaled* motion magnitude per clip as a model covariate, so the coupling score and the motion-availability covariate were distinct quantities.

3.7 Cubical persistence of video structure

We used a cubical sublevel-set filtration of $1 - A$ on the three-dimensional time-by-height-by-width array. The cubical calculation operated on the stored zero-filled volume; unsupported irregularity voxels therefore entered at the maximum filtration value rather than being treated as high-irregularity regions. This inversion makes high-irregularity voxels enter the filtration at lower threshold values. For the primary boundary convention, we placed a one-voxel layer with filtration value 1 around every face of the volume, so the added exterior cells entered only at the final threshold. Cubical persistence was calculated with giotto-tda [11], coefficients in the field of two elements, and reduced homology, which removes the single essential connected component. For every finite, positive-length H_0 interval with birth b_i and death d_i , we summed its length:

$$P_{H_0}^{\text{video}} = \sum_{i \in H_0, d_i > b_i, d_i < \infty} (d_i - b_i). \quad (1)$$

This gave one scalar per clip. Because the filtration is three dimensional, connectivity can extend across neighboring locations and neighboring times. Total persistence is a sum across threshold lifetimes, not the number of components at any one threshold and not the duration of a visible event. The video field was normalized separately for each clip, so the contrast pertains to the normalized irregularity structure under this extraction procedure.

The new methods figure makes the quantities in Eq. 1 visible. At threshold τ , a cubical cell enters when its value in $1 - A$ is at most τ (Supplementary Figure S1A). A connected region is born when newly entered cells create a component separate from those already present; its interval ends when that component merges with an older one. A birth-death plot places one retained interval at (b_i, d_i) , with lifetime $d_i - b_i$ measured vertically above the diagonal (Supplementary Figure S1B). A barcode draws the same interval as a horizontal segment from b_i to d_i (Supplementary Figure S1C). These are two displays of the *same* saved intervals. The total in Eq. 1 sums every finite positive-length interval, including the many shorter video intervals omitted from the compact barcode display.

3.8 Gaze density fields and gaze persistence

To create a gaze field on the same grid, each eligible fixation contributed its in-window duration to the temporal bins it overlapped. In each occupied time bin, that mass was spread spatially with a Gaussian kernel centered on the fixation. The kernel's standard deviation was 2.5 grid bins; it was truncated at four standard deviations and renormalized at image boundaries. If fixation k overlapped bin t for Δt_{kt} and its normalized spatial kernel was $K_k(x, y)$, the duration-weighted field was

$$G(t, x, y) = \sum_k \Delta t_{kt} K_k(x, y). \quad (2)$$

The construction was checked to conserve mass: summing the raw field over all cells reproduced the trial's eligible fixation duration. Each trial field was then divided by its own 99th percentile and clipped to $[0, 1]$. We applied the same $1 - G$ inversion, boundary padding, reduced cubical-persistence convention, and finite-interval summary as for the video field to obtain trial-level gaze H_0 total persistence. A zero total means that no finite positive-length H_0 intervals remained after reduced homology; it does not mean that the trial lacked fixations. The pipeline also constructed a point-cloud representation, but the present gaze-topology result and alignment covariate use the duration-weighted cubical field.

The gaze barcode in Supplementary Figure S1C shows all four positive-length H_0 intervals for one actual-AI T2 trial judged AI; their lifetimes sum to 2.69. The paired real and AI video barcodes show only the 12 longest intervals for each clip, while their displayed totals sum all retained intervals. Scene 06 was selected for illustration because its paired video- H_0 difference was nearest the median of the 20 scene differences. The gaze trial is the fixed example also used to illustrate location controls in Figure 2A. These displays explain the computation; they do not constitute additional tests. Per-clip video and per-trial gaze fields encode different quantities and are scaled separately, so their raw total-persistence magnitudes are not directly comparable.

3.9 Direct gaze-motion coupling and location controls

For each video time bin, the scaled motion field M was centered and divided by its spatial standard deviation over valid cells. This made a fixation's sampled value relative to the motion distribution available at that moment. We sampled the standardized field at a fixation's video-normalized location using bilinear interpolation and averaged samples over every time bin that the fixation interval overlapped, weighted by overlap duration. The primary endpoint used zero temporal lag.

The control was built separately for each fixation and feature. Radial eccentricity was computed in the physical 4:3 frame from the center using coordinates $(0.8(x - 0.5), 0.6(y - 0.5))$. Thirty angles were sampled from the circle with that radius; proposed controls outside the video or outside feature support were rejected. Every bilinear-interpolation cell touched by an observed or control point had to be valid at each relevant time bin. A fixation with no supported

overlapping time bin or with invalid observed sampling support was omitted from that trial’s coupling average; its usable fixation count and duration were recorded separately. The trial-specific random generator was seeded reproducibly from the frozen seed 20260808 and the trial identifier. Control coordinates were held fixed across the pipeline’s lag calculations; only the zero-lag result enters this report. For each of the 30 control replicates, fixation timing and duration stayed unchanged while its location was replaced by the corresponding eccentricity-matched point.

Let $z_t(x, y)$ denote the within-time-bin standardized motion field, and let w_{kt} be the seconds of fixation k overlapping time bin t . The observed trial average S and the primary direct-coupling score C were

$$S = \frac{\sum_{k,t} w_{kt} z_t(x_k, y_k)}{\sum_{k,t} w_{kt}}, \quad C = \frac{S - \bar{S}_0}{\text{sd}(S_{0,1}, \dots, S_{0,30})}, \quad (3)$$

where each $S_{0,r}$ is a whole-trial control average computed with the same temporal weights. The null standard deviation used the sample convention ($n - 1$ denominator); if it was zero, the standardized score was undefined. The score is a same-video, same-time, same-eccentricity contrast in sampled *motion magnitude*. It is not a gaze-to-flow-vector correlation and does not identify artifact-directed looking.

3.10 Conventional gaze summaries and landscape alignment

We computed three conventional summaries from the same eligible in-video fixations. Scanpath length was the sum of Euclidean distances between successive fixation centers after conversion to 960 by 720 video pixels. Dispersion was the duration-weighted root mean square distance from the duration-weighted fixation centroid, in the same pixel coordinate system. Spatial gaze entropy used a 10×10 grid: each cell’s duration share was converted to a probability p_j , and entropy was $-\sum_{j:p_j>0} p_j \log p_j$ with the natural logarithm. We retained eligible fixation count and total eligible duration separately. These summaries describe path extent, spread, spatial occupancy, and data quantity without asking whether gaze was on moving regions at the corresponding times.

The broad topological-alignment covariate used persistence *landscapes* instead of comparing raw total-persistence values. Each positive-length H_0 diagram interval contributed a triangular function over filtration threshold; at each threshold, the five largest function values were retained. Landscapes were evaluated at 100 equally spaced points on the fixed interval $[0, 1]$. The gaze-to-video distance was the square root of the trapezoidal integral of squared landscape differences, summed across these five layers. For each gaze trial, the viewed-video distance was compared with distances to 10 other S1 clips of the same actual authenticity and different scene identity, selected without replacement by a reproducible trial-specific generator. The alignment score was median alternative distance minus viewed-video distance. Positive scores indicated that the viewed video was closer to the gaze field in this topological representation than the sampled alternatives. The two fields share a grid and a $[0, 1]$ filtration range but encode different physical quantities; a landscape distance is therefore a descriptive comparison of shape, not an equality of video and gaze units. Absolute image location is not retained by this score, so its meaning differs from the direct-coupling measure in Eq. 3.

3.11 Statistical analysis

3.11.1 Matched stimulus contrast. We paired each AI clip with its corresponding real-scene clip and computed the AI-minus-real difference in video H_0 total persistence. We reported the two group means, mean paired difference, its 95% t confidence interval, and a two-sided paired t test with 19 degrees of freedom. The scene pair, not a participant’s viewing of a clip, was the independent unit. The figure shows every pair; its illustrative scene was chosen by the paired difference nearest the median, not by statistical extremity.

3.11.2 *T2 judgment model.* Mixed-effects logistic regression was fitted by maximum likelihood using `glmer` with the Laplace approximation and the `bobyqa` optimizer. An AI judgment was coded 1 and a real judgment 0. Actual authenticity was a two-level factor with real as reference. Direct coupling was standardized across the 385 eligible T2 trials and entered with an interaction with actual authenticity. Mean unscaled video motion, eligible fixation count, and eligible duration were included as covariates. These three covariates had been standardized across the full 769-trial eligible model table before the T2 subset was taken. The fitted fixed-effects expression was

$$\text{logit Pr}(J = \text{AI}) = \beta_0 + \beta_1 z(C) + \beta_2 a + \beta_3 z(C)a + \gamma_1 z(\bar{m}) + \gamma_2 z(N) + \gamma_3 z(D), \quad (4)$$

with random intercepts for participant, scene, and video. Here $a = 1$ indicates actual AI, $\bar{m}$ is the clip's mean unscaled motion, N is eligible fixation count, and D is eligible duration. Thus β_3 is the difference between the coupling slopes for actual AI and real videos on the log-odds scale; $\exp(\beta_3)$ is a ratio of odds ratios. The conditional slopes are β_1 for actual-real clips and $\beta_1 + \beta_3$ for actual-AI clips; their standard errors use the fixed-effect covariance matrix. Coefficients and conditional slopes were tested with two-sided Wald z tests, and reported odds ratios and 95% intervals exponentiate the log-odds estimates and Wald limits.

3.11.3 *T2 perception-aware topology model.* After restricting to the 385 eligible T2 trials, we examined a video- H_0 by actual-authenticity by judgment interaction in a linear mixed model. Real was the reference for both factors. The response in the saved analysis was

$$Y = \log\left(P_{H_0}^{\text{gaze}} + \epsilon_{\text{mach}}\right), \quad (5)$$

where ϵ_{mach} is the machine epsilon for double precision. Fixed predictors were standardized video H_0 , actual authenticity, judged authenticity, all two-way interactions, their three-way interaction, standardized mean video motion, eligible fixation count, and eligible duration. Unlike the judgment models, all four continuous predictors in this model were standardized within T2. The random intercepts were participant, scene, and video. The model was fitted by maximum likelihood with `lmer` and `bobyqa`; the full three-intercept fit was nonsingular. Fixed-effect tests used Satterthwaite degrees of freedom. Estimated marginal trends and their 95% intervals gave the video- H_0 slope separately for actual-real/judged-real, actual-real/judged-AI, actual-AI/judged-real, and actual-AI/judged-AI trials. The three-way coefficient tests a contrast among slopes; it is not itself the correctly detected AI slope.

Three actual-AI trials judged AI had gaze H_0 exactly zero, despite having 7, 9, and 9 eligible fixations spanning 4.87–4.91 s. Under Eq. 5, each zero is mapped to approximately -36 because $\log(\epsilon_{\text{mach}}) \approx -36$. To assess sensitivity to this transformation while retaining every observation, we refitted the identical model with $\log(1 + P_{H_0}^{\text{gaze}})$. This full-sample, zero-safe sensitivity analysis is reported in the Supplementary Material (Appendix A).

3.11.4 *Response-version sensitivity and inference limits.* The author-verified corrected judgment file was used as the primary response source. Relative to the earlier export, two actual-AI T2 judgments were corrected from “real” to “AI”: P2/participant 1 viewing set1-06-a, and P1/participant 5 viewing set1-18-a. These corrections changed two false negatives to correct detections. All gaze, video, and derived feature values were unchanged. The earlier export was retained only for provenance and sensitivity checking. All reported tests are nominal and two-sided; no multiple-testing correction was applied to this selected secondary analysis. Independent localized artifact annotations and independent plausibility ratings were unavailable, so coupling cannot be interpreted as attention to verified AI artifacts or as a causal mechanism of detection.

3.12 Software and reproducibility

The analysis used Python 3.12.7 with NumPy 1.26.4, pandas 2.3.3, SciPy 1.12.0, OpenCV 4.11.0, giotto-tda 0.6.2, and other packages recorded in stage manifests; the saved R model provenance identifies R 4.4.2. Configuration, command ledger, source checksums, stage outputs, fitted-model ladders, and input-file hashes were retained with the run. The configuration fixed the five-second window, geometry, frame rate, feature grid, flow parameters, cubical-persistence settings, 30 location controls, 10 alternative videos, and seed 20260808.

4 Results

4.1 Matched AI and real videos differed in H_0 persistence

Across the 20 matched physics scenes, mean video H_0 total persistence was 206.48 for AI clips and 374.78 for real clips. The paired AI-minus-real difference was -168.30 (95% CI $[-302.81, -33.80]$), $t(19) = -2.62$, $p = .0169$ (Figure 1). Thus, the AI clips had less total lifetime of connected components in the measured temporal-irregularity field on average; the scene-level contrasts varied in direction and magnitude. This is a video-level contrast across 20 pairs, not a participant-level gaze effect.

Supplementary Figure S2 decomposes this same paired total-persistence difference across the normalized filtration. Its area is the reported scalar contrast, rather than a separate hypothesis test. Supplementary Figure S3 separately asks how much of the contrast remains after excluding short-lived intervals; interval lifetime and position along the filtration are different properties. Supplementary Figure S4 shows the boundary-rule diagnostic for the same video contrast. None replaces the unthresholded analysis.

4.2 Direct coupling and AI judgments

Among the 385 T2 trials, 38 of 192 actual-real trials were judged AI (false alarms) and 123 of 193 actual-AI trials were judged AI (correct detections). In the adjusted all-T2 judgment model, the coupling slope differed by actual authenticity: the coupling-by-actual-AI interaction was $\beta = 1.139$ (SE = 0.363), ratio of odds ratios = 3.12 (95% CI $[1.53, 6.36]$), $p = .00168$ (Figure 2). The conditional slope for actual-real trials was $\beta = -0.550$, odds ratio = 0.58 (95% CI $[0.35, 0.94]$), $p = .0289$, indicating fewer AI judgments at higher coupling. For actual-AI trials it was $\beta = 0.589$, odds ratio = 1.80 (95% CI $[1.08, 3.00]$), $p = .0241$, indicating more AI judgments at higher coupling. These conditional estimates use the same coupling SD across all T2 trials; they describe within-sample associations, not classification accuracy.

4.3 Perception-aware video-gaze topology

The three-way interaction was $\beta = 4.624$ (SE = 1.657), $p = .00557$. The adjusted video-to-gaze H_0 slope for actual-AI clips judged AI was 4.247 (95% CI $[1.571, 6.923]$). The other three cell slopes remained imprecise or near zero (Supplementary Figure S5). Three trials in the actual-AI/judged-AI cell had gaze $H_0 = 0$; the full-sample, zero-safe transformation also yielded a nominally significant interaction, as reported in the Supplementary Material (Appendix A).

5 Discussion

This secondary analysis of Shi et al.'s eye-tracking dataset found two principal associations in its selected physics subset. AI clips had lower total persistence of connected components in the processed temporal-irregularity field than their matched real clips. Within T2 trials, the association between controlled gaze sampling of motion and an AI response differed by actual authenticity: higher coupling was associated with fewer AI judgments for actual-real clips

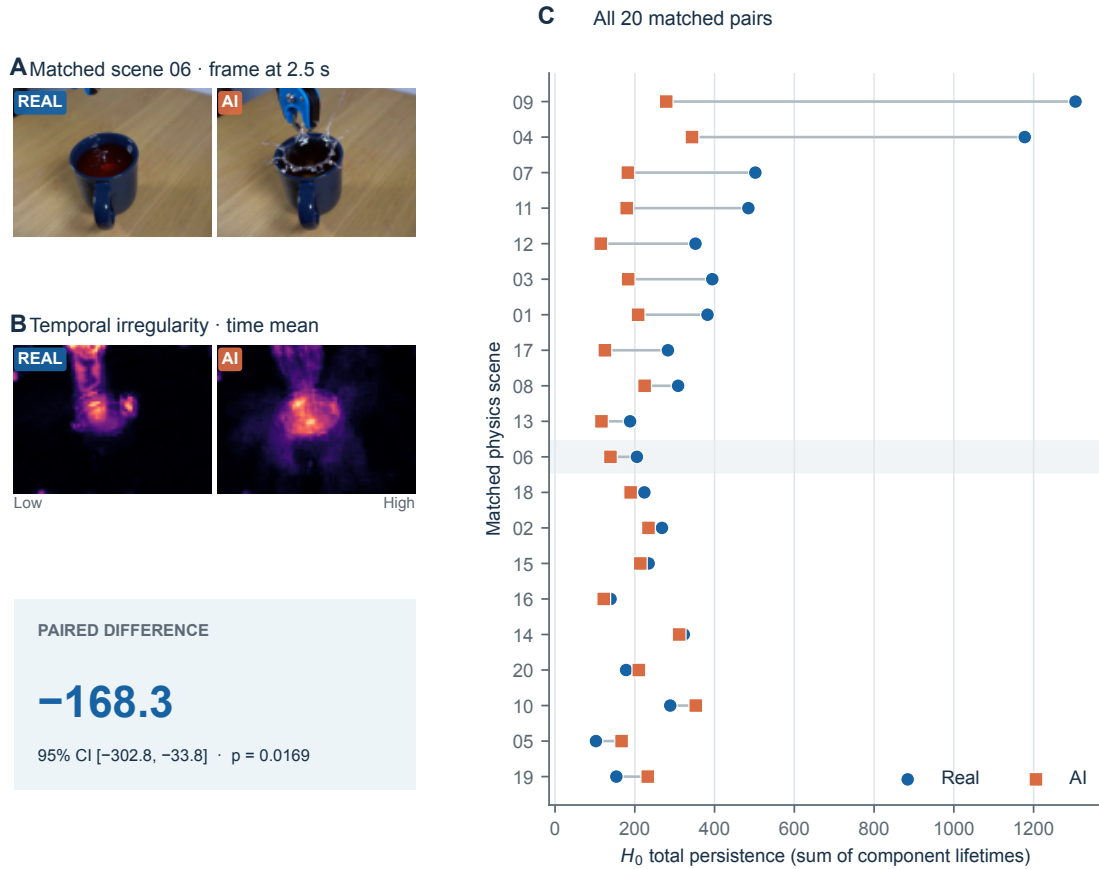


Fig. 1. **Video temporal structure in matched physics scenes.** **A**, Source-color frames at 2.5 s from matched real and AI clips in scene 06. This pair was selected because its AI-minus-real H_0 difference was nearest the median paired difference, rather than an extreme. **B**, Corresponding temporal-irregularity fields averaged across the five-second clip; the common color scale runs from low (dark) to high (bright). The fields, rather than the color frames, supplied the H_0 calculations. **C**, Each horizontal line joins real (blue circle) and AI (coral square) values for one of the 20 matched scenes; rows are ordered by paired difference. The highlighted row is scene 06. H_0 total persistence is the sum of finite connected-component lifetimes across filtration thresholds. The mean AI-minus-real difference was -168.30 (95% CI [-302.81, -33.80]; paired $t(19) = -2.62$, two-sided $p = .0169$).

and more AI judgments for actual-AI clips. The video- H_0 by actual-authenticity by judgment interaction in the gaze- H_0 model was significant under both reported all-trial outcome transformations, with weaker evidence under the zero-safe transformation. These results characterize the selected sample and processing pipeline; they do not show that gaze caused a judgment or that viewers detected a verified local defect.

The video contrast concerns the sum of connected-component lifetimes in a per-clip normalized, processed field. Its independent units are 20 scene pairs, not the 385 behavioral trials or the number of video intervals. The direction was negative in 16 pairs, and the paired median and trimmed-mean sensitivities were also negative, but three pairs (scenes 04, 07, and 09) contributed 56.7% of the sum of absolute paired differences. The no-padding computation reproduced

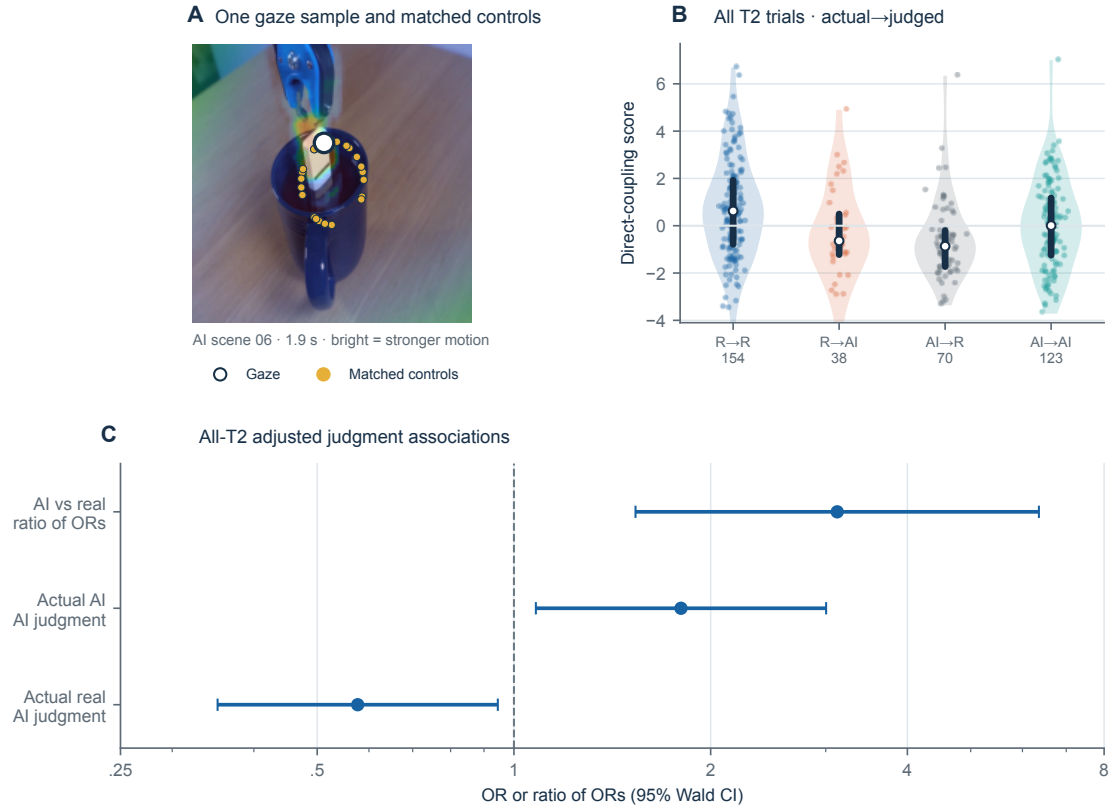


Fig. 2. Direct gaze-motion coupling and AI judgments across all T2 trials. **A**, Example AI-video frame at 1.9 s with its motion field overlaid: brighter color denotes stronger motion. The white point marks one participant's fixation; the gold points mark its 30 same-time, eccentricity-matched control locations. This panel illustrates the score construction and is not an inferential result. **B**, Direct-coupling scores for all 385 T2 trials, grouped by actual and judged authenticity using the author-verified corrected judgments. Arrows mean actual → judged; numbers are trial counts, and colors distinguish the four groups. **C**, Adjusted odds ratios for AI judgment within actual-real and actual-AI trials, and their ratio (AI versus real), with 95% Wald confidence intervals from the all-T2 model. Only the corrected judgments are plotted. Direct coupling was standardized across all 385 T2 trials. These are conditional associations, not out-of-sample discrimination measures.

the saved video H_0 and H_1 totals. Accordingly, the result is best described as a difference in measured fields for these matched clips, not as evidence that one class is more physically plausible or perceptually simpler.

The direct-coupling result is a conditional association between a trial-level measure of motion sampled at gaze locations and the participant's response. The score controlled the location comparison for time and eccentricity; the judgment model also included mean video motion and measures of gaze quantity. This design does not manipulate gaze, motion, or viewing strategy. It therefore cannot establish that sampling motion changed the response, or that the measured motion marked an anomaly. In the corrected-response sample, participants judged 38 of 192 actual-real trials as AI and 123 of 193 actual-AI trials as AI. Those counts include false alarms and misses as well as correct responses; the class-specific slopes should not be called accuracy. A separate AI-only model and in-sample likelihood comparison are reported in the Supplementary Material (Appendix A) and do not constitute external predictive validation.

The published study and this analysis share the participant cohort, source experiment, and selected physics stimuli and responses [9]. The earlier article described broad task-, authenticity-, and perceived-authenticity-related gaze summaries across a wider video library. The present analysis adds a matched video-field endpoint, an eccentricity- and time-controlled gaze-motion score, and a class-conditional association between that score and T2 judgments. A selected secondary analysis of new measures can coexist with the earlier broad results, but the shared sample means this work is not a replication.

The topology findings should be read separately. Video H_0 is a stimulus-field contrast; it is not evidence that video topology aligns with gaze topology. In the AI-only subset, adding broad landscape alignment after direct coupling did not improve the fitted likelihood. The exploratory three-way interaction had $p = .00557$ with the machine-epsilon transformation and $p = .0424$ with the full-sample, zero-safe transformation reported in the Supplementary Material (Appendix A). Topology-to-topology correspondence should also be distinguished from associations with judgments.

Several limits determine how far these conclusions travel. First, the analysis uses one selected physics category with 20 nonidentical real-AI scene pairs; pairing controls scenario identity but not frame-level content, and the pair count constrains generalization. Second, 40 participants contributed repeated trials, and T1 always preceded T2, so task and order cannot be separated. Third, these are selected secondary tests with nominal two-sided p values and no multiplicity adjustment. Fourth, optical flow, camera registration, support handling, and per-clip percentile scaling define the temporal-irregularity field; the post hoc file checks found no missing or corrupt fields but did not independently validate flow or homographies. Fifth, no independent artifact annotations or plausibility ratings were available, limiting any interpretation in terms of artifact-directed attention or physical plausibility. These constraints rule out causal, detector-performance, and external-validation claims.

A useful next study would combine a genuinely independent masked review of all clips and frame-pair flow diagnostics with a new, provenance-verified set of physics videos and participants. It should freeze video preprocessing, response reconciliation, the zero-safe gaze-topology outcome, and the model before evaluating outcomes. Claims about viewers attending to specific artifacts would additionally require independent local artifact labels; claims about detector utility would require an external prediction evaluation with a target defined as the viewer's response and features available without ground-truth class information. Jointly measuring processed stimulus structure and controlled gaze sampling offers a way to formulate sharper questions about authenticity judgments while keeping their evidential limits explicit.

6 Conclusion

This analysis adds two measures to the study of gaze during judgments of generated physics videos: a matched-pair summary of connected-component persistence in a processed temporal-irregularity field, and a control-based score of motion sampled at gaze locations. In this dataset, video H_0 persistence was lower for AI clips than for matched real clips, while the association between gaze-motion coupling and AI judgments differed across actual classes. The gaze- H_0 interaction was significant under both reported all-trial transformations, with weaker evidence under the zero-safe transformation. Because the data are reused, observational, and limited to one selected stimulus subset, the results motivate future replication.

Ethics and Privacy Statement

The original study was approved by the Department of Engineering Ethics Committee at the University of Cambridge, and participants gave informed consent, as reported by Shi et al. [9]. This secondary analysis used de-identified fixation and response records and involved no new recruitment.

Data Availability Statement

The data and analysis code supporting this study will be made publicly available in an Open Science Framework (OSF) repository at [insert OSF DOI or URL].

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A Supplementary Material

Supplementary boundary sensitivity. Using the same 40 stored video-irregularity fields, we recomputed cubical persistence with no outer padding and with a zero-valued outer layer. The no-padding calculation reproduced each clip’s primary-rule H_0 and H_1 total persistence and the mean paired AI-minus-real H_0 difference of -168.30 . Zero padding changed both totals for all 40 clips and gave a mean paired H_0 difference of -133.73 ; it is shown as a diagnostic because its exterior cells enter at the first threshold. The comparison changed only the outer boundary rule: unsupported irregularity voxels remained zero-filled, and gaze fields and video–gaze alignment were not recomputed. The plotted 0.95–1.00 threshold-band area is post hoc and descriptive; positive and negative portions of the paired Betti difference cancel within that band, so its net area does not describe the broader high-threshold region.

Supplementary scene-QC and influence analysis. After inspecting the paired result, we performed a post hoc technical review of all 40 source clips, standardized derivatives, and stored video fields. We checked source hashes, decoded frame counts and rates, field dimensions and finite values, validity-mask summaries, and consecutive-frame changes on downsampled gray frames. Masked contact sheets showed three source frames and the time-mean fields for every clip. We also inspected source, derivative, and field slices at three times for both clips in scenes 03, 04, 07, and 09. The

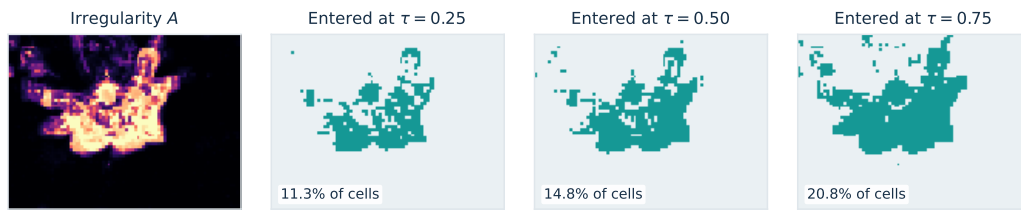
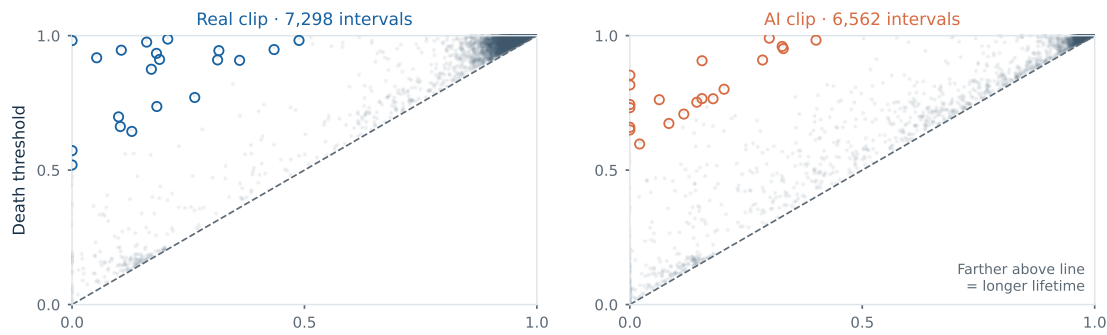
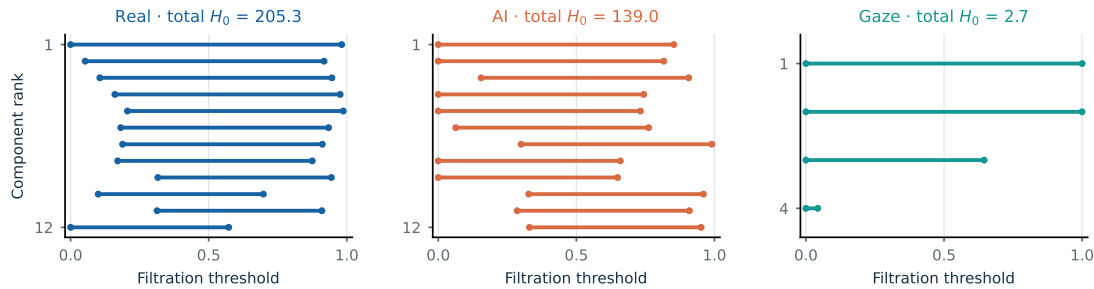
A One 2D slice through the 3D cubical field**B** Birth--death diagrams for matched scene 06**C** Barcodes · longest intervals first

Fig. S1. How cubical H_0 persistence was computed and displayed. **A**, A spatial slice at 2.54 s of the AI temporal-irregularity field in scene 06 and three thresholds of its sublevel filtration on 1 – **A**. Teal cells have entered by threshold τ ; their increasing coverage illustrates how regions appear and join. These are two-dimensional slices for explanation. The reported persistence was calculated from the complete $60 \times 60 \times 80$ time-space cubical volume, with padded boundaries. **B**, Actual H_0 birth–death diagrams from the full real and AI video volumes for matched scene 06. Horizontal position is birth threshold and vertical position is death threshold. Every finite, positive-length interval appears as a gray point; open blue or coral circles mark the 20 longest intervals. The dashed diagonal denotes zero lifetime. **C**, Barcodes draw each interval from birth to death, ordered by lifetime: the 12 longest intervals are shown for each video, and all four positive-length intervals are shown for one illustrative T2 gaze trial viewing the AI clip and judging it AI. The H_0 totals in the headings sum *all* finite positive-length intervals, including video intervals omitted from the barcode display. Video and gaze fields share the filtration range but encode different quantities, so their raw total-persistence magnitudes should not be read as directly comparable. The selected scene and gaze trial illustrate the procedure; they are not inferential units for the group contrast.

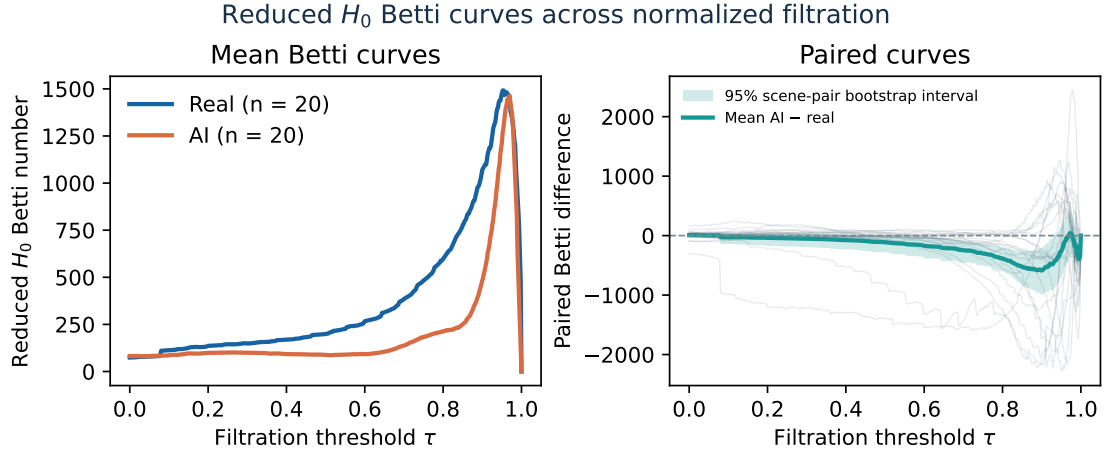


Fig. S2. **Supplementary reduced- H_0 Betti curves for the 20 matched physics-video pairs.** **A**, Mean number of active finite positive-length reduced- H_0 intervals at each threshold for real and AI clips. **B**, Paired AI-minus-real curves; faint lines show the 20 scene pairs, and the band is a percentile interval from 5,000 scene-pair bootstrap resamples. Fields were normalized separately for each clip and share the normalized $[0, 1]$ filtration scale. The exact integral of each curve equals its total persistence; the exact mean paired integral is -168.30 , the same contrast reported in the main text, rather than an independent test. Many short intervals can contribute. The plotted bootstrap interval is descriptive uncertainty for the curve; no threshold-wise significance tests were performed.

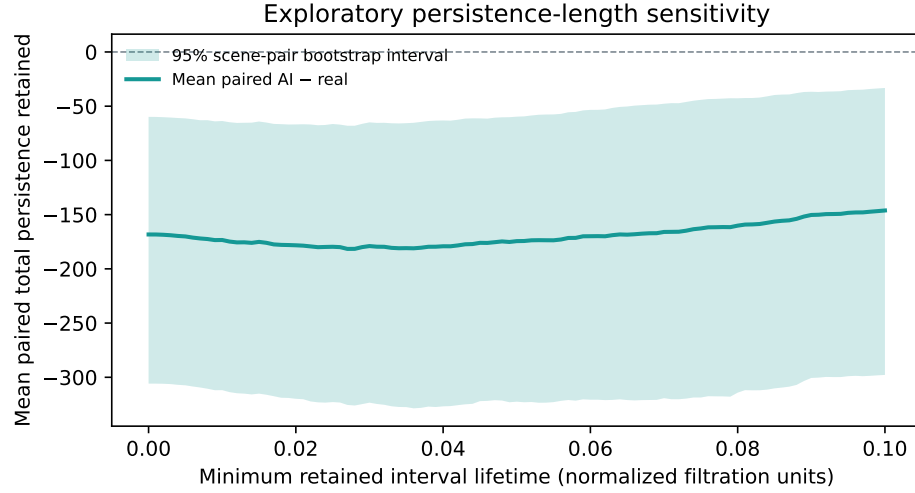


Fig. S3. **Persistence-length sensitivity for the video H_0 contrast.** At each minimum lifetime cutoff, only intervals at or above that normalized lifetime are retained before total persistence is summed for each clip. The curve shows the mean paired AI-minus-real contrast across the 20 scenes; the band shows a percentile interval from 5,000 scene-pair bootstrap resamples. The plot shows how the paired contrast changes as short-lived intervals are excluded. The unthresholded estimate is the video- H_0 result reported in the main text.

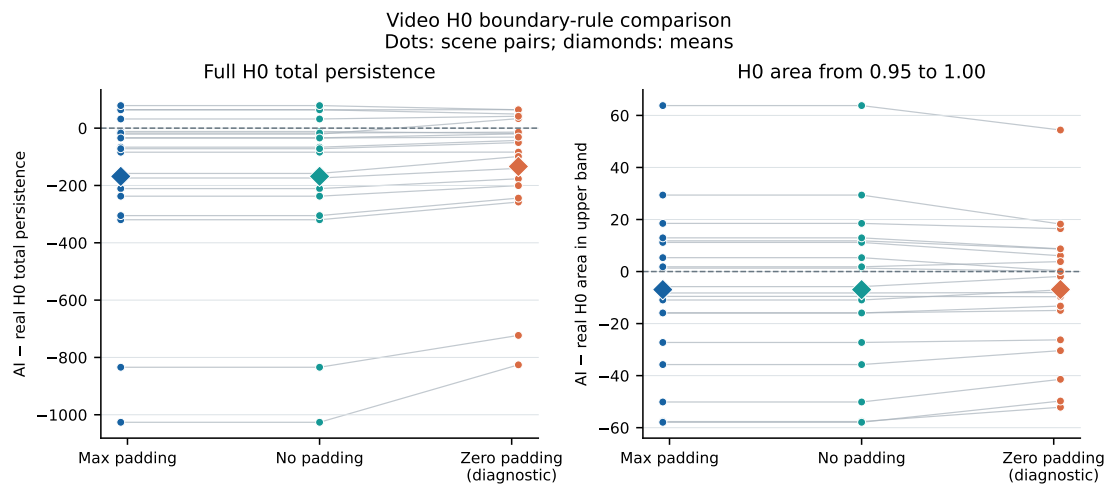


Fig. S4. **Supplementary video boundary-rule sensitivity for the 20 matched physics scenes.** Each dot is one scene’s AI-minus-real contrast; diamonds mark means, and lines connect the same scene across rules. The left panel shows full video H_0 total persistence. The right panel shows the exact net area of the paired reduced- H_0 Betti difference over the post hoc threshold band 0.95–1.00. Maximum-value padding was the primary rule. Removing the outer layer gave identical per-clip H_0 and H_1 total-persistence summaries for all 40 videos; zero padding is a diagnostic that introduces early-entering exterior cells. The sensitivity holds the saved treatment of unsupported voxels fixed and does not establish invariance of complete diagrams, gaze persistence, or video-gaze alignment. No inference is based on the post hoc threshold band.

contact-sheet codes concealed clip identity during that review, but the influential scene identities were already known; this was not an independent blinded assessment of physical plausibility or flow accuracy.

The three largest absolute paired H_0 differences (scenes 04, 07, and 09) accounted for 56.7% of the sum of absolute paired differences. The technical review found marked differences in temporal activity between some matched clips and flagged motion-field or registration concerns for further inspection. The group contrast is therefore a difference in the processed fields, conditional on this stimulus set and extraction procedure.

All 40 source hashes and decoded frame counts matched the manifest. Every standardized derivative decoded to 120 frames at 24 Hz; stored A and motion arrays were finite, and A lay in $[0, 1]$. The technical checks therefore found no missing or corrupt clip or field. They do not validate optical flow or camera registration. In particular, scene-03 AI and real clips logged 39 and 117 camera-registration failures, respectively, among 119 attempts. Scene-04 AI had mean unscaled motion 7488.9 derivative pixels/s despite zero logged registration failures, and its motion slices showed a broad high-valued region that requires flow-level review. The scene-04, -07, and -09 real clips each had more than 120 consecutive source-frame pairs with mean absolute change below 0.2 gray levels after downsampling to 64 by 48 pixels. This threshold marks near-static frames for screening; it does not imply exact duplicate frames. Per-clip percentile normalization can magnify low-amplitude residual variation, so the large persistence differences in these pairs should not be read as isolated evidence of a semantic or physical difference.

The full 20-pair contrast had median -69.01 and 20%-trimmed mean -93.32 ; descriptive scene-bootstrap 95% intervals were $[-192.39, -18.94]$ and $[-203.49, -24.65]$, respectively. Sixteen of 20 differences were negative (two-sided exact sign-test $p = .0118$). Across 20 leave-one-scene-out paired t calculations, the mean differences ranged from -181.31 to -123.15 and nominal p values from .0137 to .0306. Omitting scenes 04, 07, and 09 together gave a mean of -69.76 (95%

CI $[-127.48, -12.04]$; nominal paired $p = .0209$). These analyses were chosen after seeing the original differences and are descriptive robustness checks, not independent confirmatory tests. The review did not re-estimate optical flow or certify the scene-04 motion estimate.

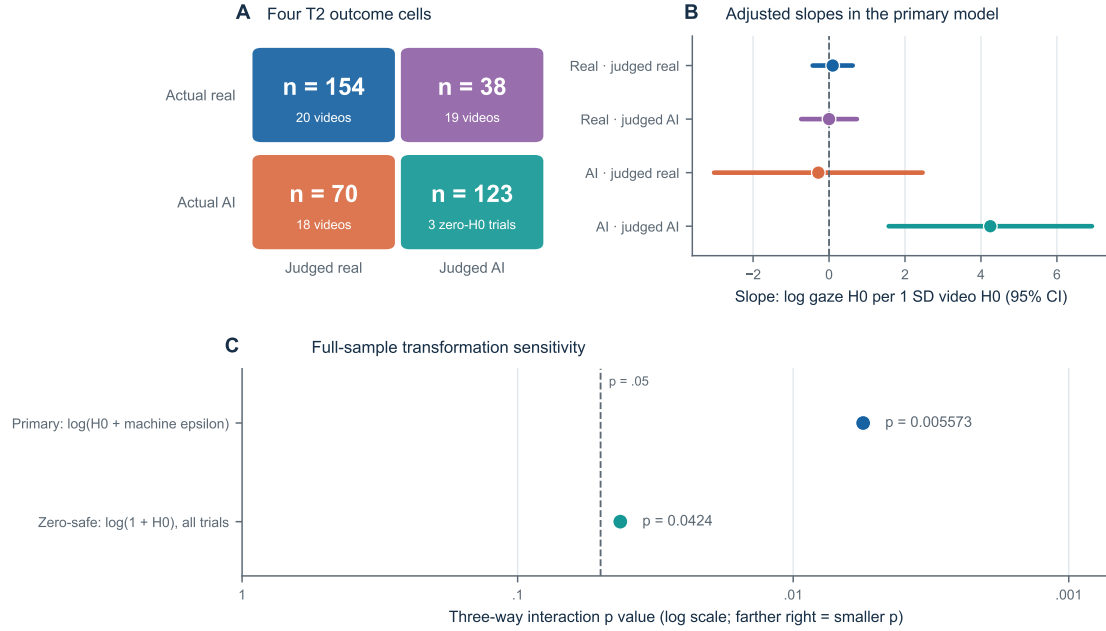


Fig. S5. **T2 video-gaze H_0 association and full-sample transformation sensitivity.** **A**, The four actual-authenticity by judgment cells using the author-verified corrected judgments (385 trials). Three of the 123 actual-AI/judged-AI trials had gaze H_0 equal to zero. **B**, Adjusted slopes of $\log(P_{H_0}^{\text{gaze}} + \epsilon_{\text{mach}})$ on standardized video H_0 , with 95% confidence intervals, from the primary corrected-judgment model. This machine-epsilon transformation maps each zero gaze- H_0 value to approximately -36 . **C**, Two-sided three-way interaction p values for the primary machine-epsilon model and the zero-safe $\log(1 + P_{H_0}^{\text{gaze}})$ sensitivity, both retaining all 385 corrected-judgment trials, including the three zeros. The dashed line marks $p = .05$.

Full-sample zero-safe transformation sensitivity. The three zero-valued gaze- H_0 observations were valid gaze trials whose reduced- H_0 diagrams contained no finite positive-length intervals, rather than missing eye-tracking records. They were retained in a sensitivity fit that replaced the primary outcome $\log(P_{H_0}^{\text{gaze}} + \epsilon_{\text{mach}})$ with $\log(1 + P_{H_0}^{\text{gaze}})$. The same fixed and random effects were used, and all 385 corrected-judgment trials remained in the analysis. The video- H_0 by actual-authenticity by judgment interaction was $\beta = 0.320$ (SE = 0.157, Satterthwaite df = 366.4), $p = .0424$. Both reported all-trial fits crossed the nominal $p < .05$ threshold, although the zero-safe fit provided weaker evidence. Coefficient magnitudes are on different response scales and should not be compared directly.

Supplementary actual-AI-only detection and incremental checks. Within the 193 actual-AI T2 trials, correct detection meant an AI judgment. A separate mixed binomial model used correct detection as outcome, direct coupling and the same three motion/gaze-quality covariates as fixed predictors, and direct coupling re-standardized within those 193 trials. A scripted fit ladder first attempted participant, scene, and video random intercepts. That fit did not meet the convergence criterion, so the accepted model used participant and scene intercepts; it was converged and nonsingular.

The coupling estimate was $\beta = 0.655$ (SE = 0.286), odds ratio = 1.93 per AI-subset SD (95% Wald CI [1.10, 3.37]), $p = .0220$. This narrower association is consistent in direction with the actual-AI conditional slope in the main all-T2 model, but the estimates use different standardization and random-effects structures.

To examine information beyond conventional gaze measures within those same AI trials, three nested detection models used the same participant and scene random intercepts. The baseline included standardized scanpath length, standardized dispersion, fixation count, standardized gaze entropy, and mean unscaled video motion. The second model added standardized direct coupling; the third additionally added standardized broad H_0 landscape alignment. Scanpath length, dispersion, entropy, coupling, and alignment were standardized within the AI subset, whereas fixation count and motion used the full-table standardization. Attempted three-intercept versions did not converge; the accepted two-intercept versions shared the same sample and random structure. Adding direct coupling improved fitted likelihood over the baseline, $\chi^2(1) = 5.01$, $p = .0252$; adding broad topological alignment afterward did not, $\chi^2(1) = 1.11$, $p = .293$. The adjusted direct-coupling coefficient in the second model was $\beta = 0.606$ (SE = 0.287), odds ratio = 1.83 (95% Wald CI [1.04, 3.22]). These are in-sample fitted-likelihood comparisons for the AI subset, not a test of prediction on new trials or a reason to infer attention to a verified artifact.